

Working with Files in Linux: sed (Stream Editor)

sed

What it does

- ◆ sed is a **stream editor** for editing text files line by line.
- ◆ Can **substitute, delete, print, or manipulate text** without opening the file in an editor.

Basic usage

```
sed [options] 'command' file
```

✓ Important Points

- ◆ Edits are **printed to stdout** by default.
- ◆ Use -i for **in-place editing** (changes the file directly).
- ◆ Supports regex for patterns.

Substitute Text

Replace text in a file.

- ◆ Replace first occurrence in each line:

```
sed 's/Nikita/Neema/' characters.txt
```

- ◆ Replace **all occurrences** in each line:

```
sed 's/Nikita/Neema/g' characters.txt
```

- ◆ Replace **in the file directly** (in-place):

```
sed -i 's/Neema/Nilam/g' characters.txt
```

- ◆ Remove text (replace with nothing):

```
sed 's/Saxena//g' characters.txt
```

Delete Lines

Remove unwanted lines from a file.

- ◆ Delete line containing a pattern:

```
sed '/Aditya/d' characters.txt
```

- ◆ Delete empty lines:

```
sed '/^$/d' characters.txt
```

- ◆ Delete specific line numbers:

```
sed '1,2d' characters.txt
```

Remove Tabs or Spaces

- ◆ Remove tabs globally:

```
sed 's/\t//g' characters.txt
```

Print Specific Lines

- ◆ Print only certain lines:

```
sed -n '12,18p' characters.txt
```

- ◆ Delete certain lines:

```
sed '12,18d' characters.txt
```

Add Newlines / Formatting

- ◆ Add blank line after each line (double spacing):

sed G characters.txt

Conditional Substitution

◆ Substitute on all lines **except** a particular line:

```
sed '28!s/Banerjee/B/g' characters.txt
```

✓ Important Points

- ◆ ! → means **not this line**
- ◆ /pattern/d → delete lines matching pattern
- ◆ s/old/new/g → replace all occurrences
- ◆ -n ... p → print only specific lines
- ◆ -i → modify file in-place

Quick Summary Table

Command	Purpose
sed 's/old/new/' file	Replace first occurrence in line
sed 's/old/new/g' file	Replace all occurrences in line
sed -i 's/old/new/g' file	Replace in file directly
sed 's/text//g' file	Remove text
sed '/pattern/d' file	Delete lines with pattern
sed '/^\$/d' file	Delete empty lines
sed '1,2d' file	Delete specific line numbers
sed 's/\t//g' file	Remove all tabs
sed -n '12,18p' file	Print only specific lines
sed '12,18d' file	Delete specific lines
sed G file	Add blank line after each line
sed '28!s/old/new/g' file	Substitute on all lines except 28